```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.arima.model import ARIMA
from sklearn.metrics import mean_squared_error, mean_absolute_error
import warnings

warnings.filterwarnings("ignore")

df = pd.read_csv(
    "/content/airline-passengers.csv",
    parse_dates=["Month"],
    index_col="Month"
)

df.index = pd.DatetimeIndex(df.index, freq="MS")

print("FIRST FIVE RECORDS")
print(df.head())

print("\nLAST FIVE RECORDS")
print(df.tail())

print("\nDATASET SHAPE")
print(df.shape)

print("\nDATASET INFORMATION")
print(df.info())

print("\nMISSING VALUES")
print(df.isnull().sum())

print("\nSTATISTICAL SUMMARY")
print(df.describe())

plt.figure(figsize=(12, 5))
plt.plot(df["Passengers"])
plt.title("Monthly Airline Passengers")
plt.xlabel("Year")
plt.ylabel("Number of Passengers")
plt.show()

result = adfuller(df["Passengers"])

print("\nADF TEST ON ORIGINAL SERIES")
print("ADF Statistic:", result[0])
print("p-value:", result[1])

if result[1] > 0.05:
    print("The series is non-stationary")
else:
    print("The series is stationary")

log_series = np.log(df["Passengers"])
```

```
log_series = np.log(df["Passengers"])

diff1 = log_series.diff().dropna()

result1 = adfuller(diff1)

print("\nADF TEST AFTER FIRST DIFFERENCING")
print("ADF Statistic:", result1[0])
print("p-value:", result1[1])

if result1[1] > 0.05:
    diff2 = diff1.diff().dropna()

    result2 = adfuller(diff2)

    print("\nADF TEST AFTER SECOND DIFFERENCING")
    print("ADF Statistic:", result2[0])
    print("p-value:", result2[1])

    d = 2
else:
    d = 1

print("\nFINAL DIFFERENCING ORDER:", d)

plt.figure(figsize=(12, 5))
plt.plot(diff1)
plt.title("Differenced Series")
plt.xlabel("Year")
plt.ylabel("Differenced Values")
plt.show()

train_size = int(len(df) * 0.8)

train = df["Passengers"].iloc[:train_size]
test = df["Passengers"].iloc[train_size:]

print("\nTRAINING DATA SIZE:", len(train))
print("TESTING DATA SIZE:", len(test))

model = ARIMA(
    train,
    order=(2, d, 2)
)

model_fit = model.fit()

print("\nARIMA MODEL SUMMARY")
print(model_fit.summary())

forecast = model_fit.forecast(
    steps=len(test)
)

forecast.index = test.index

print("\nFORECASTED VALUES")
```

```
print(forecast)

mse = mean_squared_error(
    test,
    forecast
)

mae = mean_absolute_error(
    test,
    forecast
)

rmse = np.sqrt(mse)

print("\nEVALUATION METRICS")
print("MSE :", mse)
print("MAE :", mae)
print("RMSE:", rmse)

plt.figure(figsize=(12, 6))

plt.plot(
    train,
    label="Training Data"
)

plt.plot(
    test,
    label="Actual Test Data"
)

plt.plot(
    forecast,
    label="Predicted Data"
)

plt.title("ARIMA Airline Passenger Forecast")
plt.xlabel("Year")
plt.ylabel("Number of Passengers")
plt.legend()
plt.show()
```

```
FIRST FIVE RECORDS
            Passengers
Month
1949-01-01         112
1949-02-01         118
1949-03-01         132
1949-04-01         129
1949-05-01         121

LAST FIVE RECORDS
            Passengers
Month
1960-08-01         606
1960-09-01         508
1960-10-01         461
1960-11-01         390
1960-12-01         432

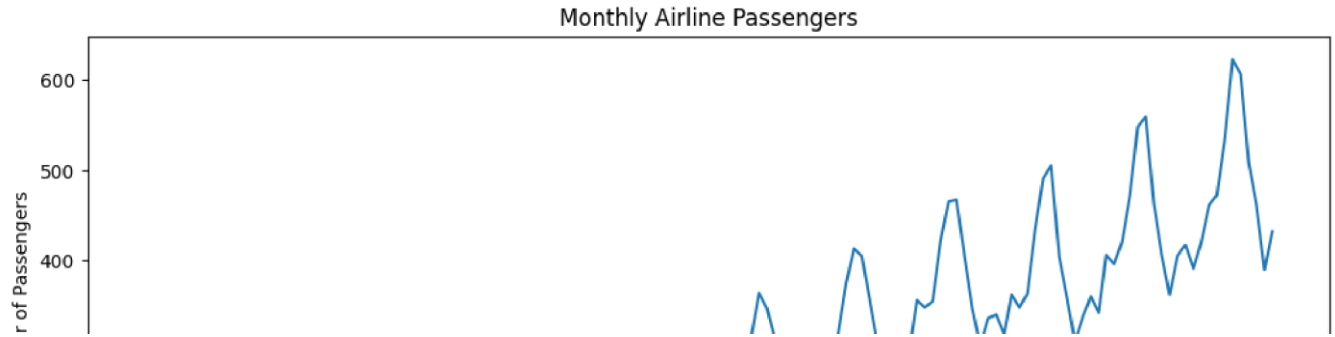
DATASET SHAPE
(144, 1)

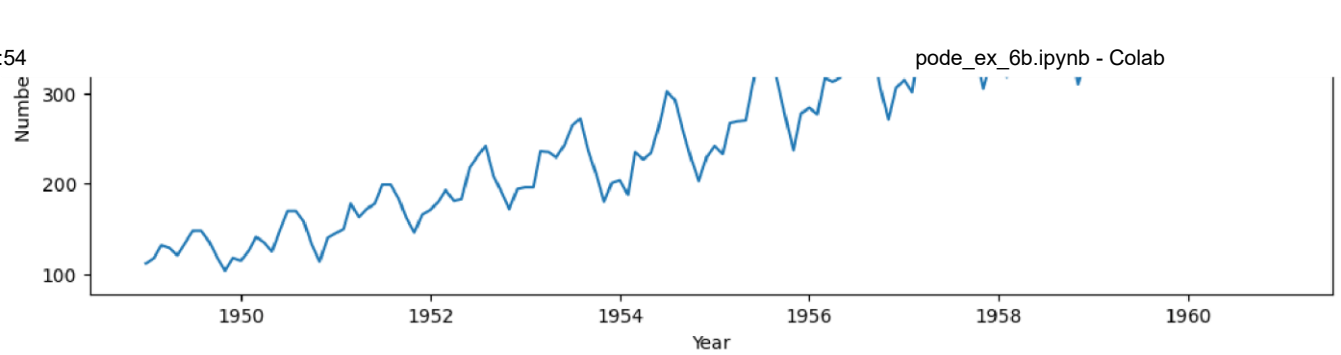
DATASET INFORMATION
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 144 entries, 1949-01-01 to 1960-12-01
Freq: MS Data columns (total 1
columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  ----
0   Passengers  144 non-null    int64
dtypes: int64(1) memory usage: 2.2 KB
None

MISSING VALUES
Passengers    0
dtype: int64

STATISTICAL SUMMARY

Passengers  count
144.000000   mean
280.298611    std
119.966317    min
104.000000    25%
180.000000
50%    265.500000
75%    360.500000
max    622.000000
```

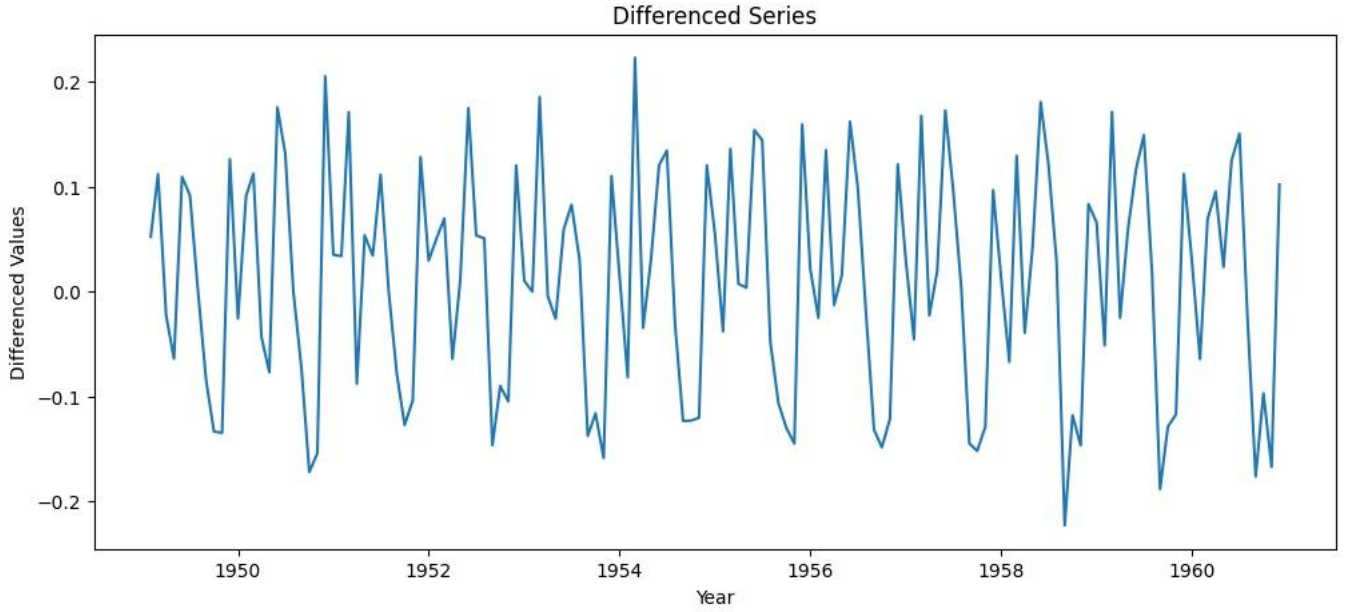





```
ADF TEST ON ORIGINAL SERIES
ADF Statistic: 0.8153688792060498
p-value: 0.991880243437641 The
series is non-stationary

ADF TEST AFTER FIRST DIFFERENCING
ADF Statistic: -2.717130598388114 p-
value: 0.07112054815086184

ADF TEST AFTER SECOND DIFFERENCING
ADF Statistic: -8.19662913218224 p-
value: 7.419304549376186e-13 FINAL
DIFFERENCING ORDER: 2
```




```
TRAINING DATA SIZE: 115
TESTING DATA SIZE: 29

ARIMA MODEL SUMMARY
                               SARIMAX Results
==============================================================================
```

```
Dep. Variable:             Passengers   No. Observations:                  115
Model:                 ARIMA(2, 2, 2)   Log Likelihood                -523.084
Date:                Wed, 30 Sep 2026   AIC                           1056.169
Time:                        04:39:49   BIC                           1069.806
Sample:                    01-01-1949   HQIC                          1061.702
                         - 07-01-1958
Covariance Type:                 opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.5036      0.122     -4.143      0.000      -0.742      -0.265
ar.L2          0.0912      0.099      0.921      0.357      -0.103       0.285
ma.L1         -0.0552     24.247     -0.002      0.998     -47.578      47.468
ma.L2         -0.9447     22.900     -0.041      0.967     -45.828      43.939
sigma2       587.1939   1.43e+04      0.041      0.967   -2.73e+04    2.85e+04
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):                 2.25
Prob(Q):                              1.00   Prob(JB):                         0.32
Heteroskedasticity (H):               5.15   Skew:                             0.11
Prob(H) (two-sided):                  0.00   Kurtosis:                         2.34
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).

FORECASTED VALUES
Month



1958-12-01    521.992754
1959-01-01    524.876742
1959-02-01    528.737670
Dep Variable:              Passengers    No Observations:                  115
1958-08-01    508.392539
1958-09-01    509.653321
1958-10-01    515.516648
1958-11-01    517.590598



1959-03-01    531.968117
1959-04-01    535.605194
1959-05-01    538.979977
1959-06-01    542.523947
1959-07-01    545.958786
```

```
1959-08-01    549.464018
1959-09-01    552.923844
1959-10-01    556.412958
1959-11-01    559.883181
1959-12-01    563.365589
1960-01-01    566.840137
1960-02-01    570.319755
1960-03-01    573.796103
1960-04-01    577.274560
1960-05-01    580.751657
1960-06-01    584.229631
1960-07-01    587.707039
1960-08-01    591.184813
1960-09-01    594.662350
1960-10-01    598.140040
1960-11-01    601.617632
1960-12-01    605.095287
Freq: MS, Name: predicted_mean, dtype: float64

EVALUATION METRICS
MSE : 18381.742370702606
MAE : 120.27019224229898
RMSE: 135.5792844453112
```

